

Yuheng Zhu

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Education

North Carolina State University <i>Ph.D. in Computer Science</i>	<i>Aug 2024 – Present</i>
North Carolina State University <i>Master in Computer Science</i>	<i>Aug 2022 – May 2024</i>
Southern University of Science and Technology <i>B.E. in Computer Science and Technology</i>	<i>Sept 2017 – May 2021</i>

Experience

Machine Learning Engineer Intern <i>Waymo</i>	<i>Mountain View, CA</i> <i>May 2026 - Aug 2026</i>
Engineer Intern <i>Qualcomm</i>	<i>San Diego, CA</i> <i>Jun 2025 - Aug 2025</i>
Research Assistant <i>North Carolina State University</i>	<i>Raleigh, NC</i> <i>Sept 2023 - Present</i>
Research Assistant <i>Southern University of Science and Technology</i>	<i>Shenzhen, China</i> <i>June 2021 – July 2022</i>
Software Engineer Intern <i>JD.com, AI Research Center</i>	<i>Chengdu, China</i> <i>June 2020 – Aug 2020</i>

Publications

AdaptAV: Continuous Adaption of Vision Models for Autonomous Vehicles Using Cloud-based Oracle. *Yuheng Z.*, Dhruva U., Boluo G. & Man-Ki Y., *Proceedings of the 100th IEEE Vehicular Technology Conference*

Bridging Data and Knowledge: A Neurosymbolic Framework for Reliable Network Analysis. Zhjin Y., *Yuheng Z.*, Mingzhe C. & Yuchen L., *Proceedings of 2025 IEEE Global Communications Conference*

FrameScope: Temporal Data Valuation for Stream Active Learning in Autonomous Vehicle Systems. *Yuheng Z.*, & Man-Ki Y. *Proceedings of The 10th ACM/IEEE Symposium on Edge Computing*

RinneFormer: Transformer-based Real-world Cooperative Perception Algorithm. Deveshwar H., *Yuheng Z.*, Dhruva U., Man-Ki Y. & Seth H. *Proceedings of 2026 IEEE Intelligent Vehicles Symposium*

Projects

E2E Learnable Association Multi-Object Tracking Model	<i>Waymo, 2026</i>
- Designed and implemented an E2E learnable-association model for 3D multi-object detection and tracking, built the full detection-to-tracking pipeline in JAX from perception backbone through association to track management. - Built a multi-frame parallel scheme with fully compilable static-shape to boost training and inference efficiency, injected 3D geometric priors into the association module for robust matching, exploring the performance and efficiency of next-generation perception world models for autonomous driving.	
Transformer-based Cooperative 3D Object Detection	<i>NCSU, 2025</i>
- Designed a Transformer-based V2X detection framework featuring Point Transformer and Cross Attention modules to capture fine-grained geometry within pillars to resolve sparse LiDAR geometry, achieving SOTA accuracy and superior generalization in real-world cooperative datasets. - Engineered a distributed ROS multi-agent pipeline robust to asynchronous data; validated system deployability and latency stability using raw sensor streams from physical autonomous vehicles.	
Agentic LLM Testcase Triage Framework	<i>Qualcomm, 2025</i>
- Independently built an agentic RAG testing failure analysis pipeline for 10K+ tests/day. - Multi-source logs distilled via LLM summarization + error-anchor context, then chunked and indexed in a vector storage (ChromaDB). - Orchestrated with LangChain agents + function calling for hybrid retrieval (semantic + keyword) and top-k reranking, emitting JSON-schema reports. - Tools used: LangChain, ChromaDB, FAISS	
On-the-fly Coding of Vision Inputs for Evidence-Preserving Perception	<i>NCSU, 2023 - 2024</i>
Designed and implemented a visual perception middleware for autonomous driving systems that preserves	

complete and reproducible evidence of visual inputs.

- Utilized compression techniques such as JPEG and H.264, with performance optimizations achieved through NVENC and NVDEC hardware acceleration engines.
- Tools used: **C, Python, V4L2, , NVIDIA Jetpack, NVIDIA Codec, FFmpeg**

Carla1s Distributed Online Simulation Platform

SUSTech, 2021 - 2022

- Designed and developed a Carla-based remote and hardware-free web simulation platform.
- Developed a **backend streaming server** for **HTTP Streaming ROS** Image Topics and Carla Sensor Output, supporting multiple video encodings and streaming protocols, reduce latency of remote simulation from 1-2s to 50ms.
- Tools used: **Python Flask, FFmpeg, Ansible, Git, Docker**

Jingxiaozi AI Customer Service System

JD.com, 2020

- Participated in two development cycles (2.1.6 & 2.1.7) of JD Jingxiaozi online AI customer service system, made in-depth research on the application and practice of Chinese **NLP** in e-commerce scenarios
- Developed web API automated testing modules in **Java** based on JD JSF **RPC framework**, made test framework can fully mimic user operation logic, increased API test coverage to 100%
- Tools used: **Java, RESTful API, JavaScript, Redis, RabbitMQ**

Technologies

Languages: C, C++, Shell, Python, Java, JavaScript, Lua

Skills: Linux Kernel, ROS2, Networking, DevOps, Docker, Redis, Carla, ADAS, LangChain, ChromaDB